§1. Background.

Defn. A Kleinian group Γ is a discrete subgp. of $PSL(2,\mathbb{C})$.

$\Gamma < PSL$ has two natural actions:

- by isometries on $\mathbb{H}^3$. Γ Kleinian implies action is discontinuous, so $\mathbb{H}^3/\Gamma$ is a hyperbolic 3-orbifold.
- by conformal actn of $S^2 = \partial\mathbb{H}^3 = \hat{\mathbb{C}}$. This action is not discontinuous, even for Γ discrete. Let:

$\Omega(\Gamma)$ = set of $x \in \hat{\mathbb{C}}$ s.t. Γ is discont. at x

$\Lambda(\Gamma)$ = set of limit points of orbits of Γ

$\overset{\text{(Thm.)}}{=} \hat{\mathbb{C}} \setminus \Omega(\Gamma)$.

$\Omega(\Gamma)/\Gamma$ is a (poss. empty, poss. disconnected) Riemann surface. Can view

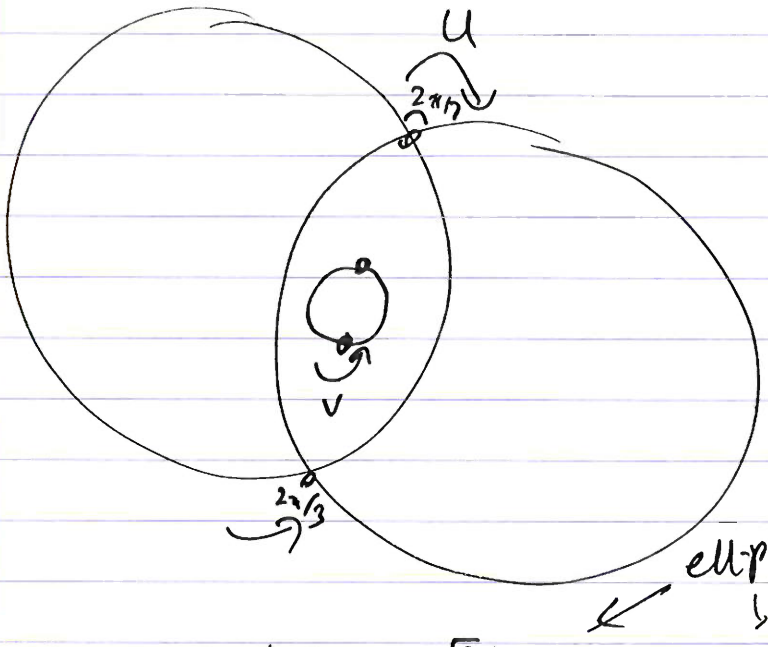
$$\Omega(\Gamma)/\Gamma = \partial\, \mathbb{H}^3/\Gamma$$

but possibly there are ends of $\mathbb{H}^3/\Gamma$ which do not admit a conformal structure in the way.

Ahlfors: Γ fin. gen. $\Rightarrow$ $\Omega(\Gamma)/\Gamma$ finite type

<u>but</u> $\mathbb{H}^3/\Gamma$ may be "infinite type" in sense of not admitting a finite-sided fundamental polyhedron in $\mathbb{H}^3$.

Egs.

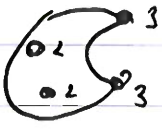


ellipt.

$$\Gamma = \left\langle u = \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix},\ v = \begin{pmatrix} 0 & -2 \\ 1/2 & 0 \end{pmatrix} \right\rangle$$

when $|\lambda| >> 0$:

- $\Gamma \simeq \mathbb{Z}/3\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$
- $\Omega(\Gamma)/\Gamma =$

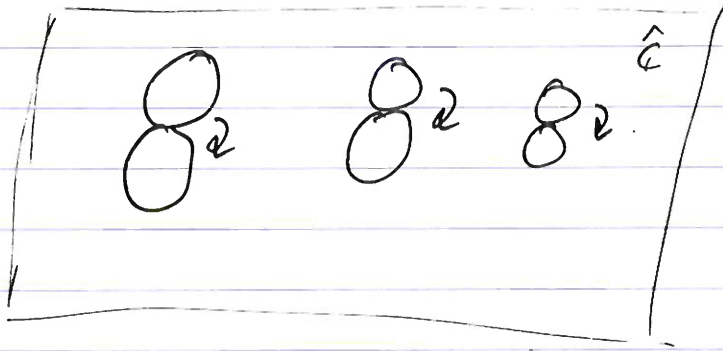


- $\mathbb{H}^3/\Gamma =$



orb. fold w/ 2 ideal singular arcs, $(\mathbb{H}^3/\Gamma) \setminus \text{Sing} \cong$ genus 2 handlebody.

Eg.



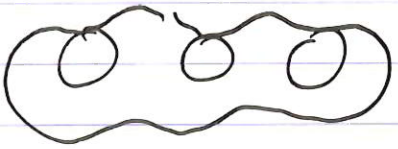
3 parabolic generators, Schottky type.

e.g. set $T = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$

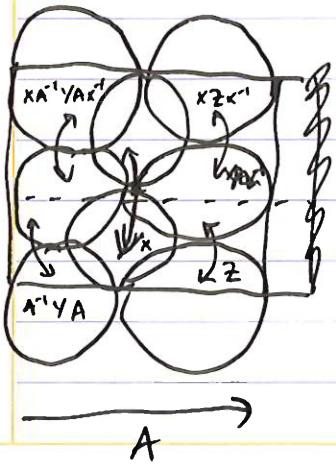
$S = \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix}$

$\Gamma = \langle S,\ TST^{-1},\ T^2ST^{-2} \rangle.$

$\Gamma \simeq \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$

$\mathbb{H}^3/\Gamma =$ 

Eg.



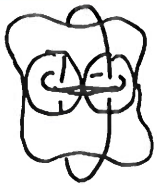


$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

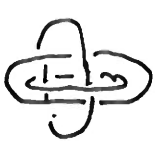
$X = \begin{pmatrix} 1 & 0 \\ -4i & 1 \end{pmatrix}$

$Y = \begin{pmatrix} -3i & \frac{3}{2}+2i \\ -4i & 2+3i \end{pmatrix}$

$Z = \begin{pmatrix} i & -\frac{1}{2} \\ 4i & -2-i \end{pmatrix}.$

$\mathbb{H}^3/\Gamma =$ 

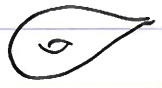
2 rank 2 parabolics,
3 rank 1 parabolics.

c.f.  = Borromean rings 

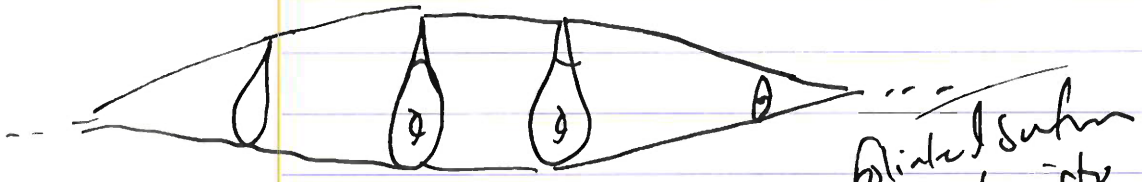
E.g. of a geom. infinite gp: (Jørgen. Maskin).

Set $c = \frac{1}{2} - i\frac{\sqrt{3}}{2}$, then

$\Gamma = \left\langle \begin{pmatrix} c & c \\ c & 1 \end{pmatrix}, \begin{pmatrix} 2-2c & -1 \\ -1 & c \end{pmatrix} \right\rangle$.

$\mathbb{H}^3/\Gamma$ =  $\times$ $(0,1)$.

2 ends, both geom infinite.



Thm (Ahlfors-Bers-Kra). If $\Omega(\Gamma)/\Gamma \neq \emptyset$, then Γ admits deformations parametrised by a quotient of $\mathrm{Teich}(\Omega(\Gamma)/\Gamma)$

i.e. $\forall\ \sigma \in \mathrm{Teich}(\Omega(\Gamma)/\Gamma)$ $\exists$ a deformation $\tilde{\Gamma}$ of Γ s.t. $\Omega(\tilde{\Gamma})/\Gamma = \sigma$,

where $\tilde{\Gamma} = f \Gamma f^{-1}$ for some q.c. auto^{m} of $\hat{\mathbb{C}}$.

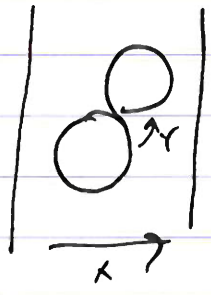
Qn What is the shape, inside $\mathrm{Hom}(\Gamma, PSL(2,\mathbb{C}))$, of the space of these deformations, $\mathcal{DH}(\Gamma)$?

§2. Ping-pong results.

Brenner (1955). When $|z|>4$,

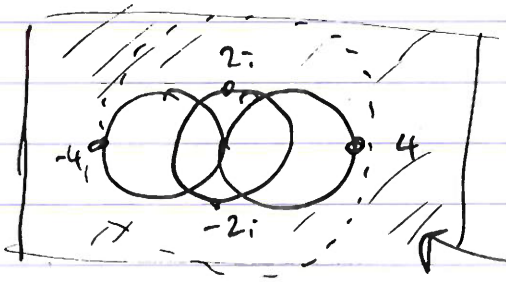
$$G_z = \langle \underbrace{\begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}}_{X}, \underbrace{\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}}_{Y} \rangle \text{ is discrete \& free.}$$

Proof:



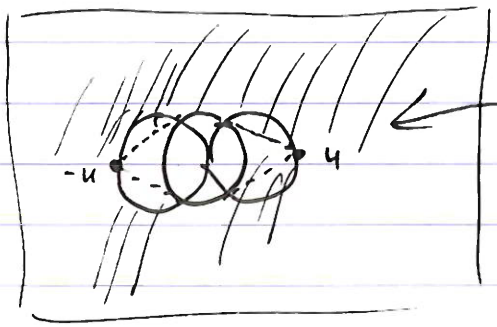
When the dom. circle of Y lie inside the strip of width 1, this is a fundamental domain. Thus $|z|>4$.

Chang-Jering-Ree (1958).



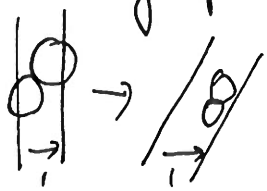
discrete + free when z lies outside the 3 circles of rad 2.

Lyndon-Ullman (1999)



exterior of convex hull of central circle with I_4.

Essentially, modify the group by changing the f.d. for X:

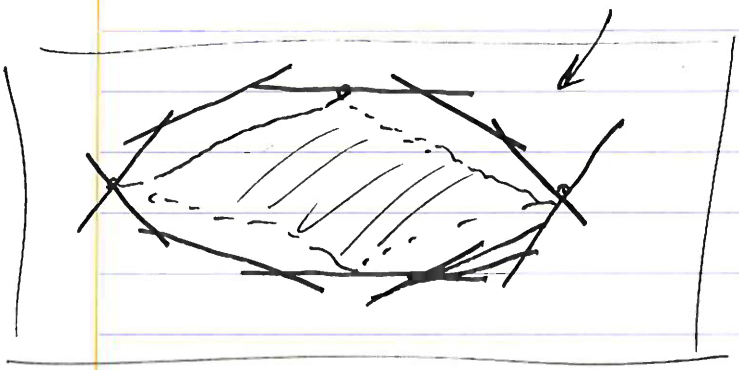


E. Gong-Martin-Schillewaert, to appear AIF '26:
- We generalize to case of 2 elliptic or parabolic generators, of orders $p \& q$
- Blindly generalizing L-U gives suboptimal results if $p \neq q$. So we use asymmetry to further reduce the region of "unknown reps".

Thm. $\exists$ a decagon $\Omega_{p,q} \subset \mathbb{C}$ s.t. if $z \notin \Omega_{p,q}$ then

$$\left\langle \begin{pmatrix} e^{i\pi/p} & 1 \\ 0 & e^{-i\pi/p} \end{pmatrix}, \begin{pmatrix} e^{i\pi q} & 0 \\ z & e^{-i\pi/q} \end{pmatrix} \right\rangle$$

ρ discrete & $\ker \rho^n$ to $\mathbb{Z}/p\mathbb{Z} * \mathbb{Z}/q\mathbb{Z}$.



Applications. (i) (in progress): enumeration of all arithmetic subgroups of $PSL(2,\mathbb{R})$ generated by two elliptic gens. If bds on deg of trace field are known, $\exists$ only finitely many candidate gps, in each cpct set, $\Omega_{p,q}$ ρ such a cpct set.

(ii) A conj. of Maxer-Gerard, Osin-Kahn, & Vesnin: The specialized Burau rep^n

$$B_3 \xrightarrow{t \neq 0} \left\langle \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix} \right\rangle$$

ρ faithful outside $\frac{3-\sqrt{5}}{2} \leq |t| \leq \frac{3+\sqrt{5}}{2}$.

Bands of this form are "semialg. nbds of ∞" in the Teichmüller space. They do not approximate $QH(G)$ well.

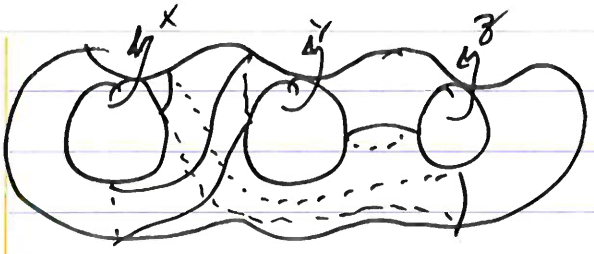
§3. Semialgebraic approximation to creeps.

Thm. (E. '25?): Let G be a geom. finite Kleinian group with $\Omega(G) \neq \emptyset$. Then $QH(G)$ decomposes as a (countable, principle) semialg. set in $\mathrm{Hom}(a, PSL(2,\mathbb{C}))$.

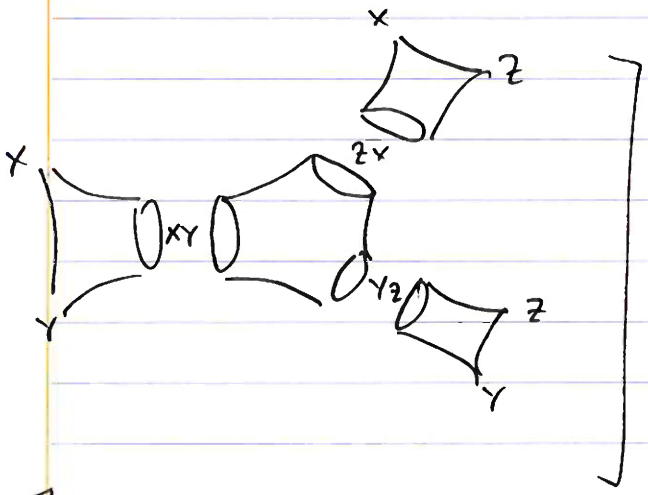
I sketch the construction in the special case "$\Omega(G)/G$ is a single $2n$-punctured sphere".

Step 1. Fix a basepoint $G \in QH(G)$ s.t. the pleating locus of $\partial(\text{conv } \Lambda(G)/G)$ is a rational lamination. Then $\Lambda(G)$ cuts out a pattern of overlapping round circles.

E.g.

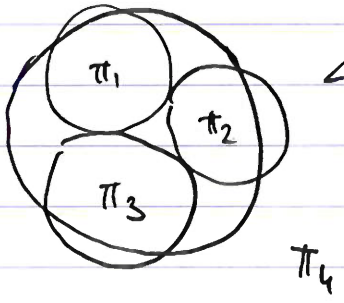


Λ is represented by XY_1, Y_2Z, & ZX.



decomposition of ∂ hyperbolic Λ/G into flat pieces.

Then $\Lambda(G_0)$ cuts out



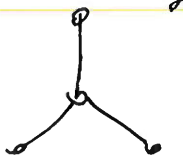
together with translates of this pattern by ~~the~~ scc. on $\Omega(G)/G_0$ transverse to the fixed Λ.

Somehow this pattern is a f.d for $G \curvearrowright \Lambda(G_0)$. It's called a "circle chain".

STEP 2. Draw a f. d. for each flat piece cut out by Ponzlin circles for each Fuchsian gp.



(Ends with gaps give technical problem about "matching up" around cycles. Here the graph is a tree so no problem.)



STEP 3. Observe, circles in $\hat{\mathbb{C}}$ are parametrized by $\mathbb{RP}^3(\mathbb{R})$ and pairs of circles are embedded $\mathbb{RP}^1$'s.
$\Rightarrow$ we can encode the fundamental domain as a combinatorial object in $\mathbb{RP}^3(\mathbb{R})$.

In particular, we can define an algebraic map on trace coordinates which gives us back the fundamental domain for G_0.

This map is algebraic, so it extends across $\mathrm{Hom}(G_\Lambda, PSL)$. In particular, for all groups with ideal lamination Λ we get a well-defined fundamental domain

$$\Phi_\Lambda(G_0) \subset (\mathbb{RP}^3)^N.$$

* when the ends have gens, no longer true: need arbitrarily many maps to cover the ideal vertices.

STEP 4. The map Φ_Λ extends uniquely to $\mathrm{Hom}(G, PSL)$ & $\exists$ an open semialg. subset where the circles have the same combinatorics in $\hat{\mathbb{C}}$ as those in the image of gps with ideal lam. Λ. This gives fundamental domains for all gps in a semialg. nbd. of the lam. of Λ-ideal lam. gps.

Since the set of groups with rational ploidy loci is dense in $\mathbb{Q}H$, the union of all these regions across all Λ gives the desired semialg. decomposition. ☒.

Rmks.
- Some technicalities involved in passing from $(\mathbb{P}^3)^N$ to actual fundatal domains, related to "twisting".
- Actually constructing the semialg. sets is possible but requires computationally intensive work in the curve complex (trace computation).

Qns for future work.
- We are essentially giving fundamental domains for the geometrically finite ends. Seems to be possible to extend to a f.d. of the whole group, except that instead of the pleated ("shrink-wrap") surfaces used by Minsky etc. pleated surfaces need to be used, unclear what they are. (ie. can couple the groups, not the conds. at the surfaces.)
- Can any of the choices be made canonical? Lots of 'flex' in all constructions.
- Our constr. is very similar to one found in Blayac, Hamenstädt, Marty (2025) but with conformal data not hyperbolic data.